5-Axis Desktop Robot Arm — Assembly Guide

Six STS3215 12 V (C018) bus servos · R5B wrist + TPU-pad gripper · ArmDojo

Printed parts: 21 PLA + 2 TPU 95A
AD16 x2; AD17 and AD20 both needed
Optional: AD28 camera platform

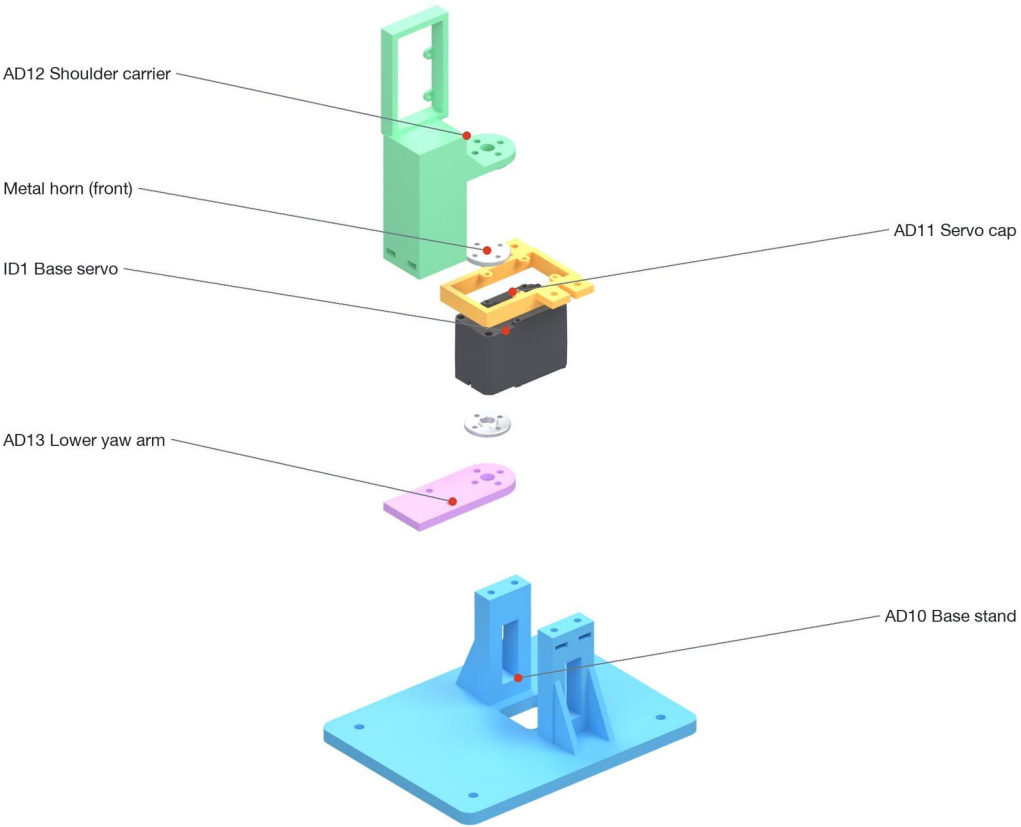
Screws (no washers):
M3x6 x4 M3x8 x40
M3x10 x6 (2 with camera)
M3x12 x29 (33 with camera)
M3 hex nuts x35 (5.5 AF, 2.4 mm)
Servo: 11 metal horns, 6 centre screws, 24 case screws
Base: 4x M4, 104 x 76 mm

Servo IDs: 1 base, 2 shoulder, 3 elbow, 4 wrist pitch, 5 wrist roll, 6 gripper

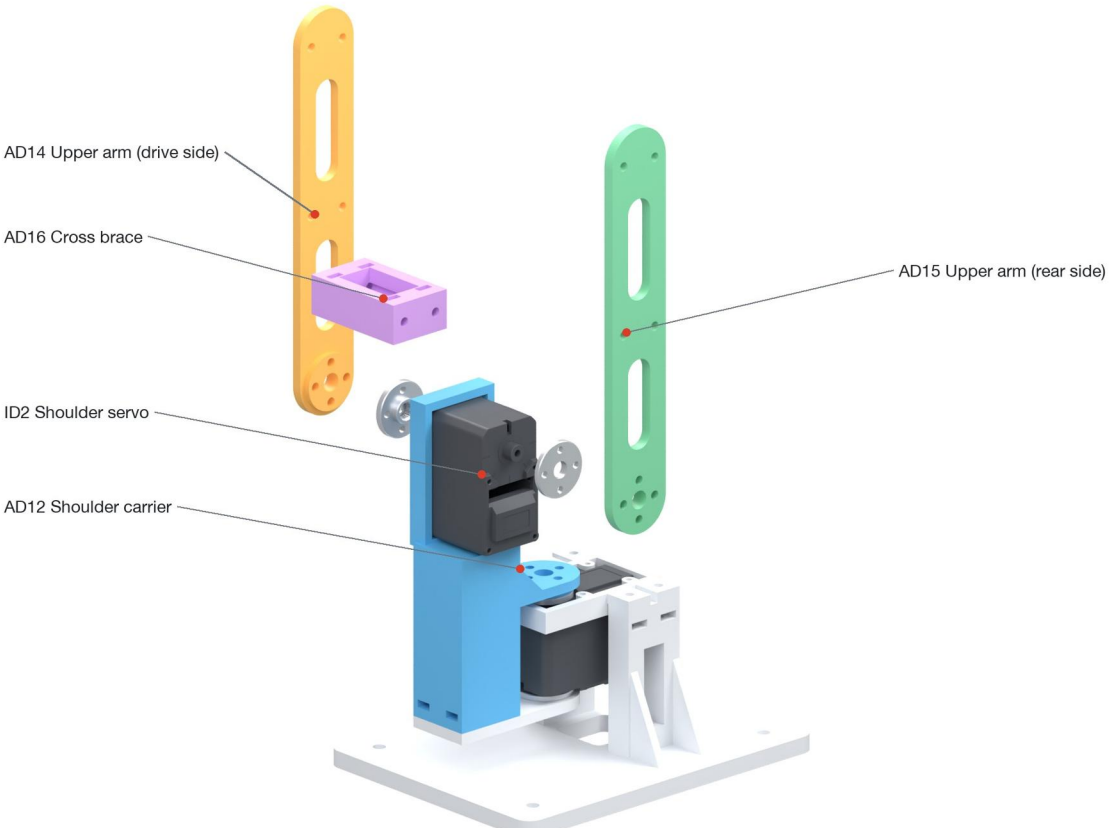
12 V servos only. Low speed and torque limit for first tests.



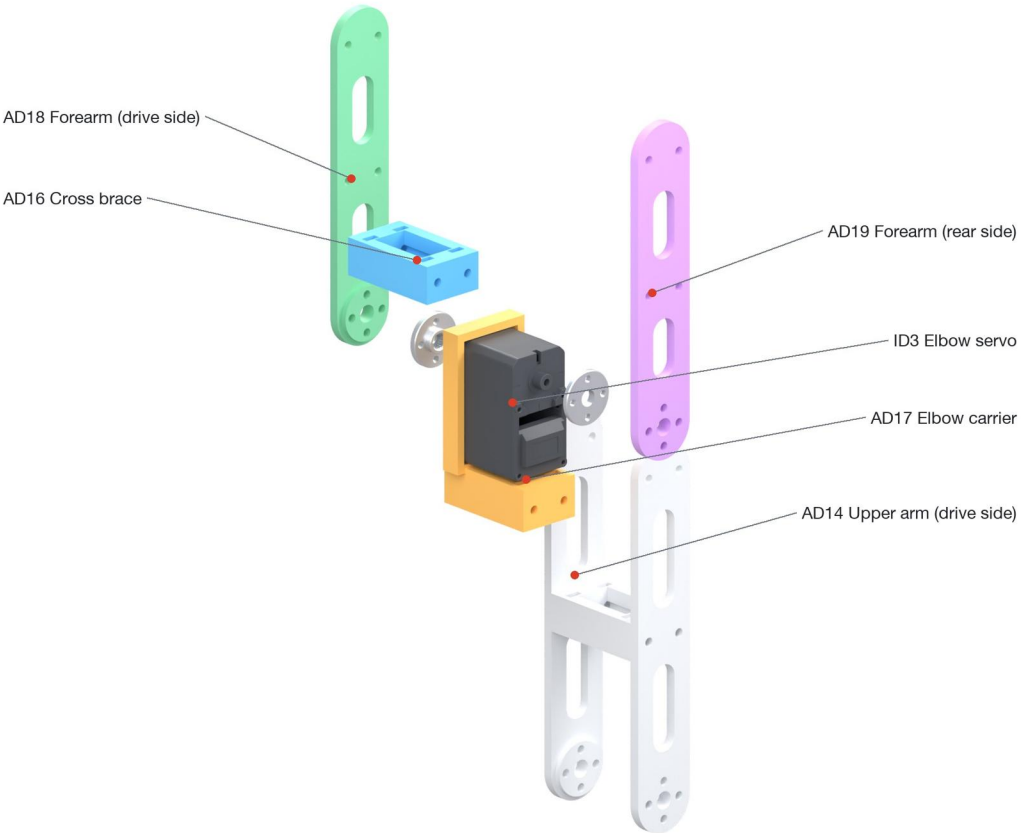
Step 1 — Base joint (ID1)



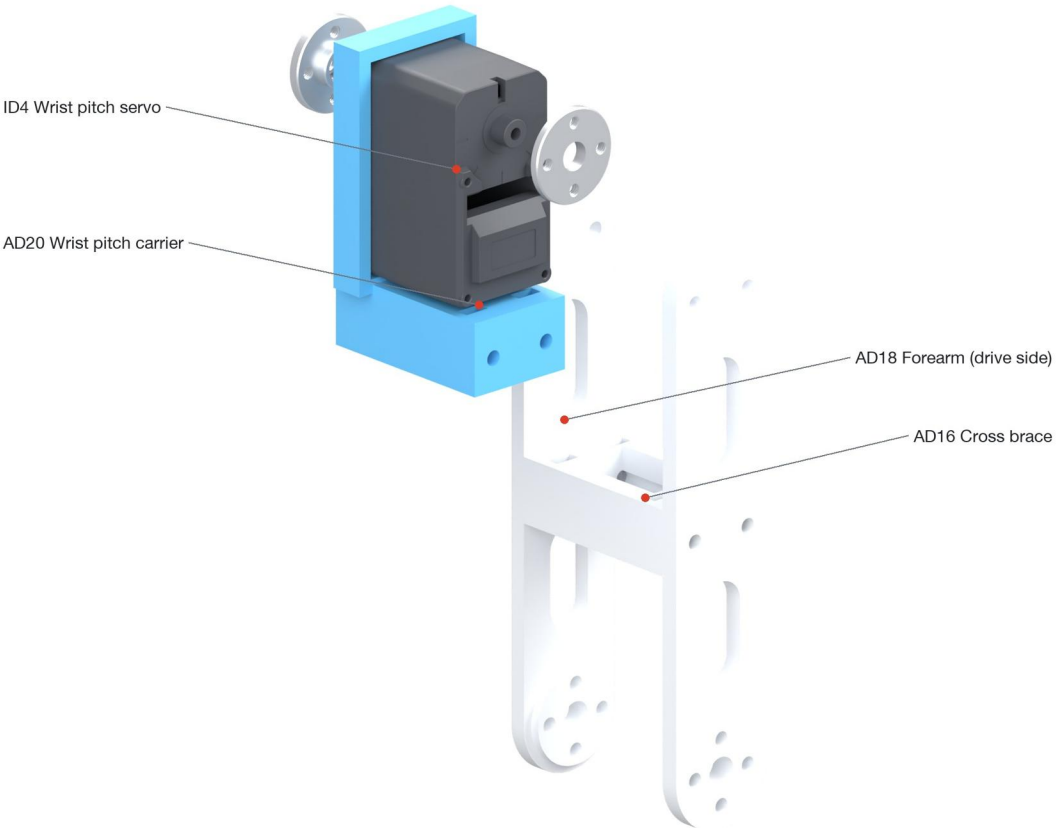
Step 2 — Shoulder (ID2) and upper arm



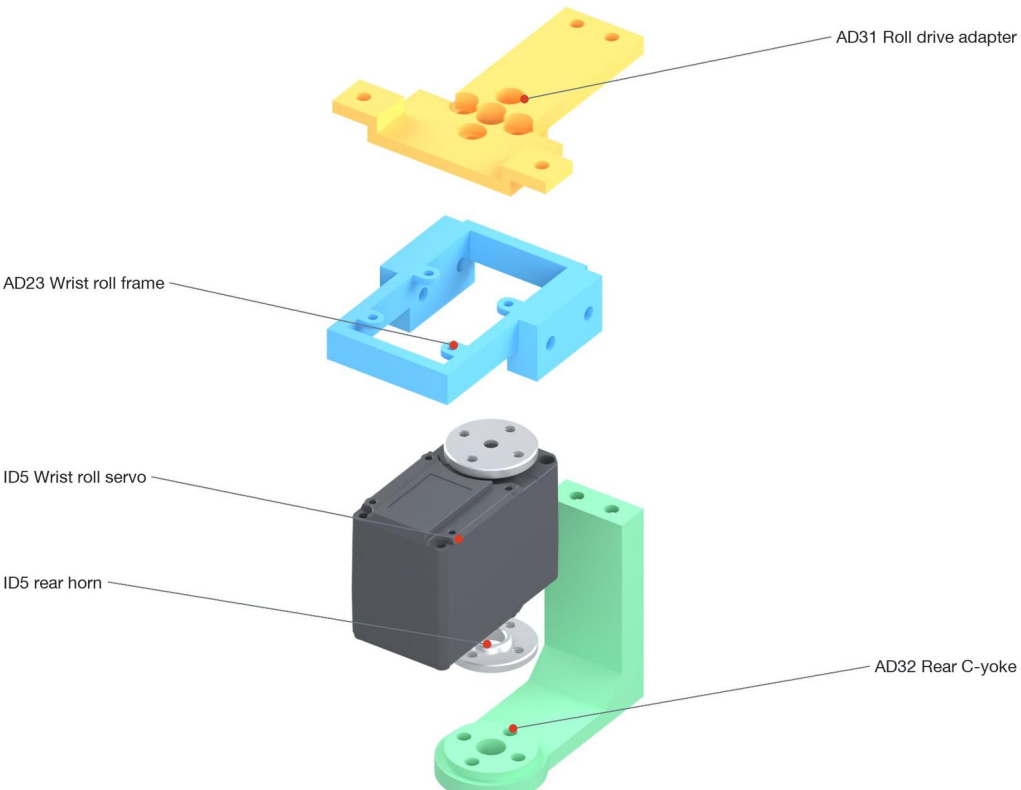
Step 3 — Elbow (ID3) and forearm



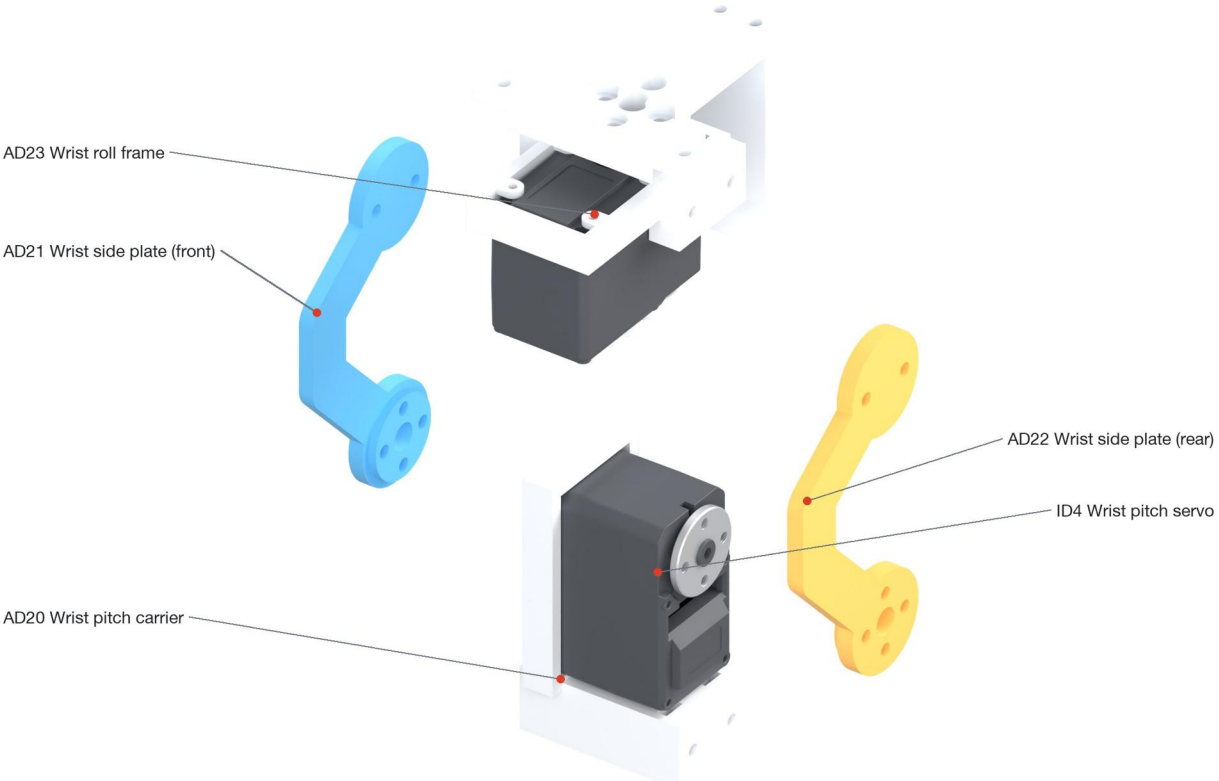
Step 4 — Wrist pitch carrier (ID4)



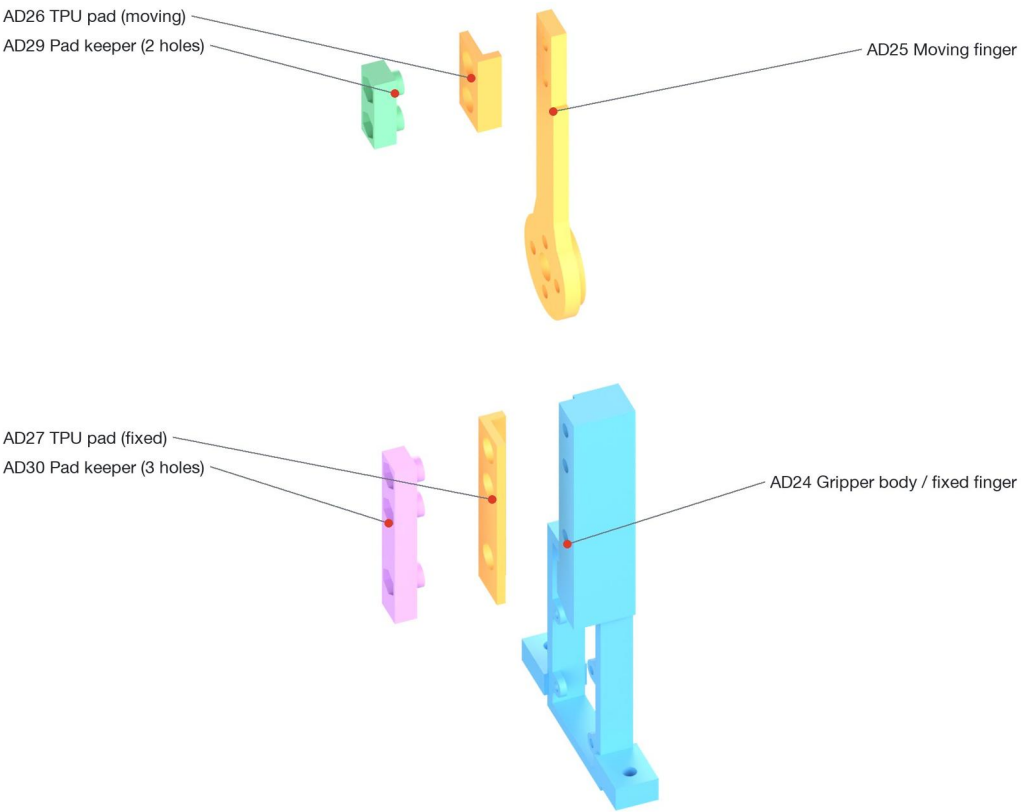
Step 5 — Wrist roll (ID5) with front adapter + rear C-yoke



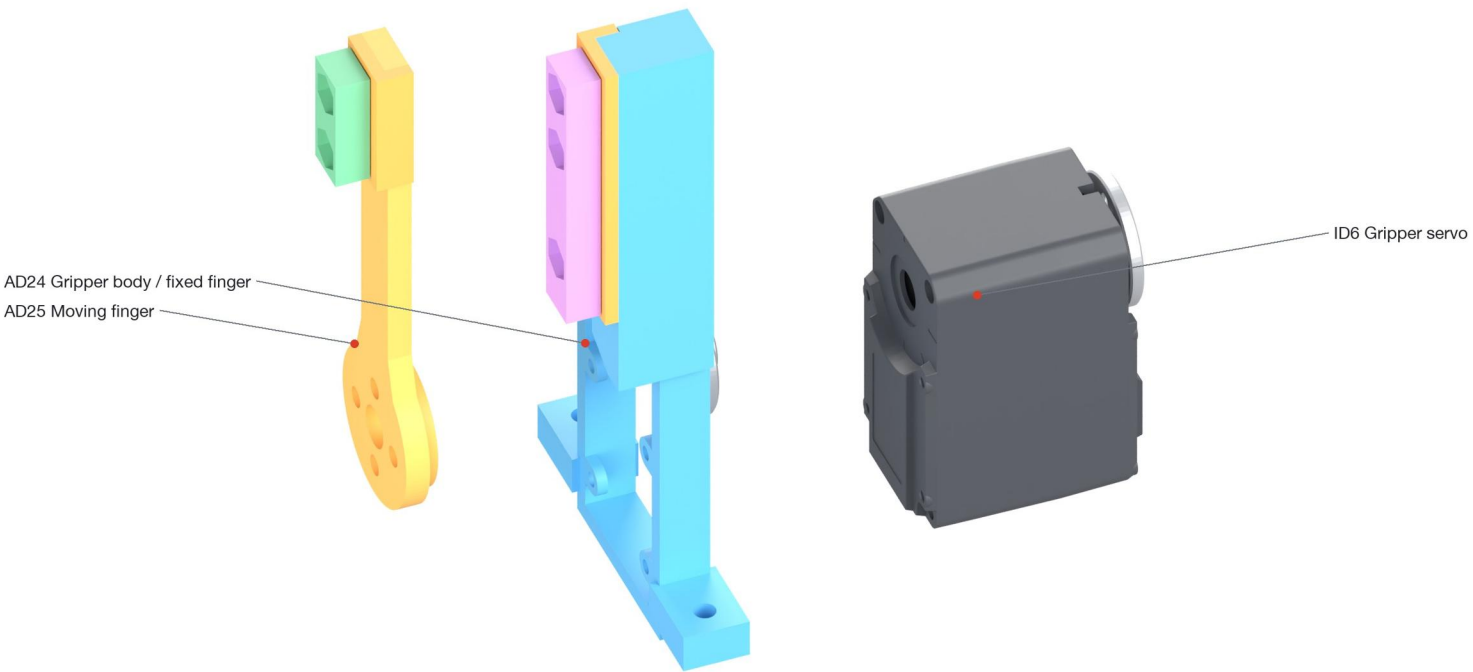
Step 6 — Side plates join wrist pitch (ID4) to wrist roll



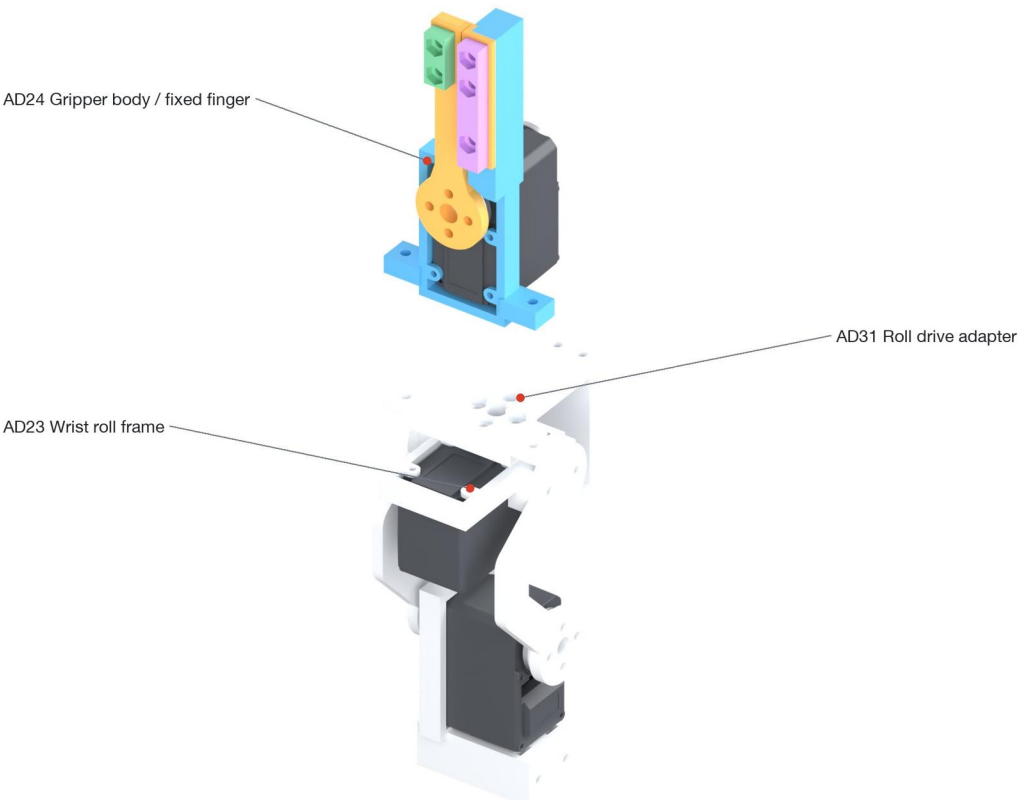
Step 7 — TPU pads and keepers on both fingers



Step 8 — Gripper servo (ID6) and moving finger



Step 9 — Gripper module onto the roll adapter (2 × M3×12)



Step 10 (optional) — Wrist camera platform

